

CoCo 2 PMODE vs. SCREEN

<https://gemini.google.com/app/3aabc2fbcb7f112>

User prompt: On a coco2, what is the difference between `PMODE 4, 1` and `SCREEN 1, 1` and what do the numbers mean?

Response: On the Tandy Color Computer 1 and 2 running Extended Color BASIC, **PMODE** controls where and how BASIC draws graphics in memory, while **SCREEN** controls what the video hardware actually outputs to your television or monitor.

Because these functions are decoupled, you can draw to a graphics buffer off-screen using PMODE before switching the display with SCREEN, or implement page flipping without visible flicker.

PMODE 4, 1 (Drawing Setup)

Syntax: PMODE <mode>, <start_page>

- **4 (Mode):** Selects the graphics resolution and color depth. Mode 4 is the highest two-color resolution available on the CoCo 2: **256 × 192 pixels** in 2 colors. This mode requires 6,144 bytes of RAM (4 memory pages).
- **1 (Start Page):** Specifies the starting video memory page in RAM where subsequent drawing commands (LINE, CIRCLE, DRAW, PSET, PCLS) will write. CoCo video memory is divided into 1.5 KB (1,536-byte) blocks called pages, beginning at RAM address &H0E00 (page 1). Because Mode 4 requires 4 pages, specifying page 1 allocates pages 1 through 4 (&H0E00 to &H25FF) as the active drawing canvas.

SCREEN 1, 1 (Display Switch)

Syntax: SCREEN <screen_type>, <colorset>

- **First 1 (Screen Type):** Determines whether the Motorola 6847 video display generator (VDG) displays text or graphics.
 - 0 = Text mode (standard 32 × 16 alphanumeric screen).
 - 1 = Graphics mode (switches the display from text to the high-resolution graphics memory).
- **Second 1 (Color Set):** Selects the hardware palette for the screen. For two-color modes like PMODE 4:
 - 0 = Color Set 0: **Black background with green foreground.**
 - 1 = Color Set 1: **Black background with white/buff foreground.**

(Note: In 4-color modes such as PMODE 1 or 3, Color Set 0 provides Green/Yellow/Blue/Red, while Color Set 1 provides Buff/Cyan/Magenta/Orange.)

To see a walkthrough of configuring graphics buffers and plotting points in Extended Color BASIC, check out [Programming BASIC on the Color Computer: PSET Pixels](#). This video covers navigating

Extended Color BASIC manuals, initializing graphics modes, and drawing onto the CoCo's screen buffer.

http://googleusercontent.com/youtube_content/1

Sources

1. [The 9 colors of the CoCo's high resolution screens](#)
 2. [Programming BASIC on the Color Computer Chapter 14 - PSET Pixels](#)
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User prompt: What are the comparable keywords on a coco3 for hi-res graphics?

Response: The Color Computer 3 replaces the old decoupled PMODE and SCREEN workflow with dedicated **Super Extended Color BASIC** commands prefixed with **H** (for High Resolution), driven by the CoCo 3's custom **GIME** (Graphics Interrupt Memory Enhancement) chip.

The primary keyword replacing both PMODE and SCREEN is **HSCREEN**, supported by **PALETTE** and **HCOLOR**.

1. HSCREEN (Mode Selection & Display Switch)

Unlike the CoCo 1/2 where you set a memory buffer with PMODE and separately enabled the video generator with SCREEN, **HSCREEN <mode>** configures both the GIME resolution/depth and immediately makes that screen visible.

Command	Resolution	Colors	Notes
HSCREEN 0	Text Mode	—	Exits graphics and returns to the standard text screen.
HSCREEN 1	320 × 192	4	4 colors selectable from the 64-color palette.
HSCREEN 2	320 × 192	16	CoCo 3's most popular mode (16 simultaneous colors).
HSCREEN 3	640 × 192	2	Highest horizontal resolution in monochrome.
HSCREEN 4	640 × 192	4	High-resolution with 4 simultaneous colors.

(The GIME hardware technically supports 200- and 225-line heights and 16-color 640-pixel modes through direct register programming/MMU banking, but Super Extended Color BASIC exposes these standard 192-line modes directly).

2. PALETTE (Color Assignment)

Instead of the CoCo 2's rigid hardware color sets (SCREEN 1, 0 vs SCREEN 1, 1), the CoCo 3 uses fully programmable palette registers.

- **PALETTE <slot>, <color>**

- `<slot>`: The logical color register used in drawing routines (e.g., 0 to 15 in HSCREEN 2).
- `<color>`: Any hardware color code from the **64-color** master palette (0–63).
- **PALETTE RGB vs PALETTE CMP**: Sets the default palette mapping to look correct on either an analog RGB monitor or a composite/TV display.

3. The H Drawing Suite

All legacy drawing commands have direct H-counterparts in Super Extended Color BASIC that operate within the active HSCREEN:

CoCo 1 / 2 Command	CoCo 3 Replacement	Function
COLOR <code><fg></code> , <code><bg></code>	HCOLOR <code><fg></code> , <code><bg></code>	Sets the active foreground and background palette slots.
PCLS <code><color></code>	HCLS <code><color></code>	Clears the active graphics screen to a palette slot.
PSET(<code>x</code> , <code>y</code> , <code>c</code>) / PRESET	HSET(<code>x</code> , <code>y</code> , <code>c</code>) / HRESET	Plots or unplots a pixel coordinate.
LINE (<code>x1</code> , <code>y1</code>) - (<code>x2</code> , <code>y2</code>)	HLINE (<code>x1</code> , <code>y1</code>) - (<code>x2</code> , <code>y2</code>)	Draws lines or filled rectangles (B / BF).
CIRCLE (<code>x</code> , <code>y</code>), <code>r</code>	HCIRCLE (<code>x</code> , <code>y</code>), <code>r</code>	Draws ellipses and circles.
PAINT (<code>x</code> , <code>y</code>), <code>c</code> , <code>b</code>	HPAINT (<code>x</code> , <code>y</code>), <code>c</code> , <code>b</code>	Flood-fills bounded areas.
DRAW "string"	HDRAW "string"	String-driven vector drawing.
GET / PUT	HGET / HPUT	Blits graphic sprite buffers (paired with HBUFF).
(None)	HPRINT (<code>x</code> , <code>y</code>), "text"	Prints rendered text directly onto the pixel canvas.

(Note: The CoCo 3 retains complete backward compatibility—running PMODE and SCREEN still executes original VDG modes identically to a CoCo 2).

Sources

1. [Tandy Service Manual - Color Computer 3](#)
 2. [CoCo III Secrets Revealed Edition 1 1986](#)
 3. [The 9 colors of the CoCo's high resolution screens](#)
 4. [Coco III Secrets Revealed \(Microcom Software\)](#)
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